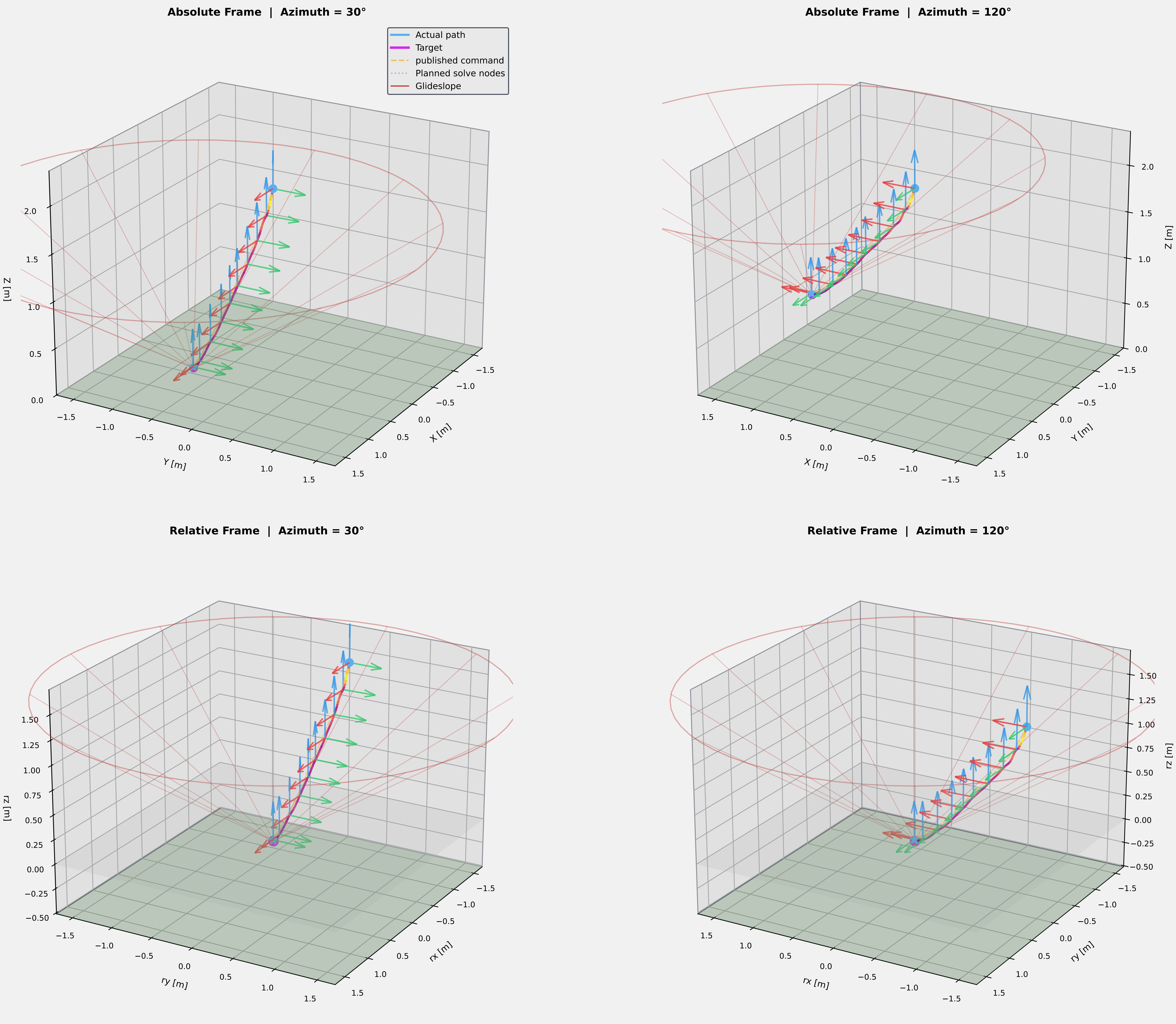
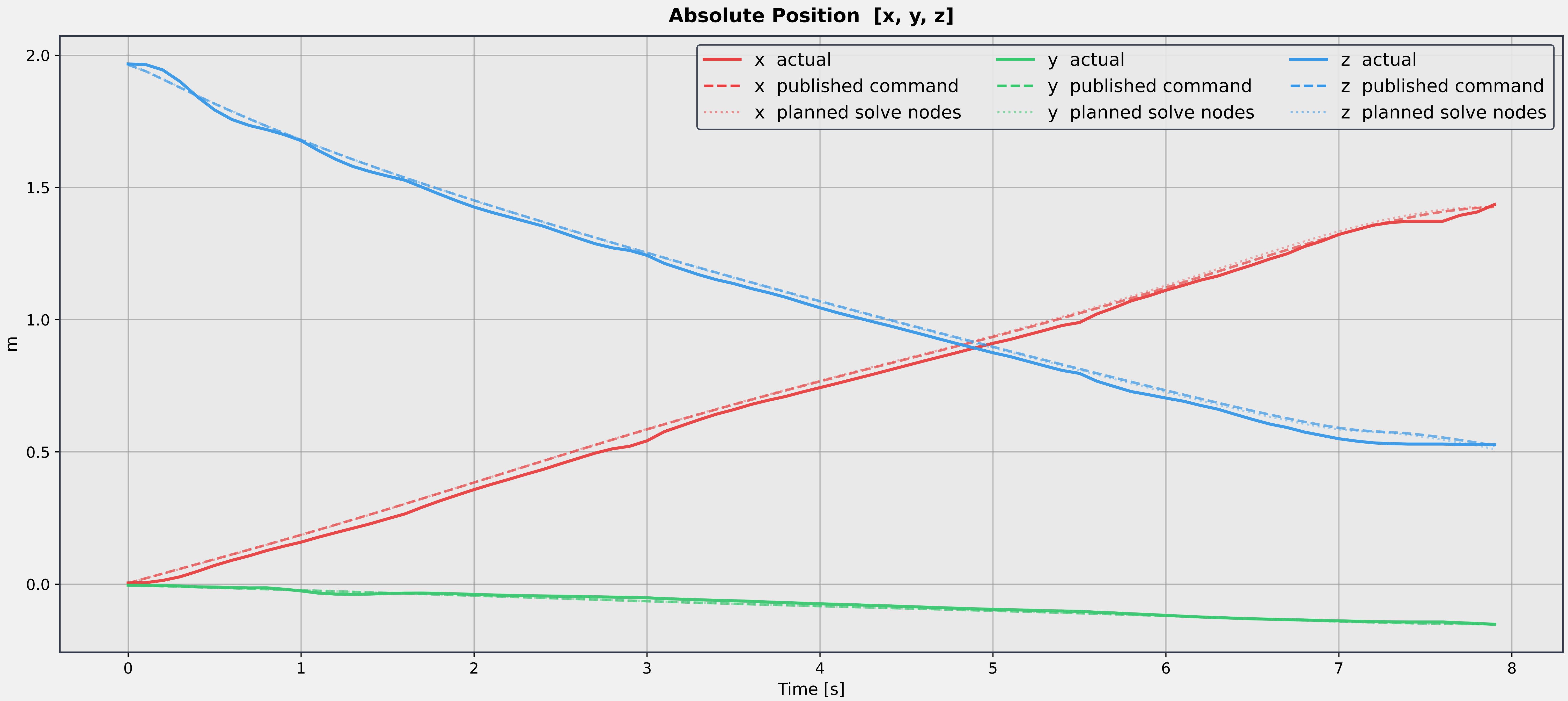
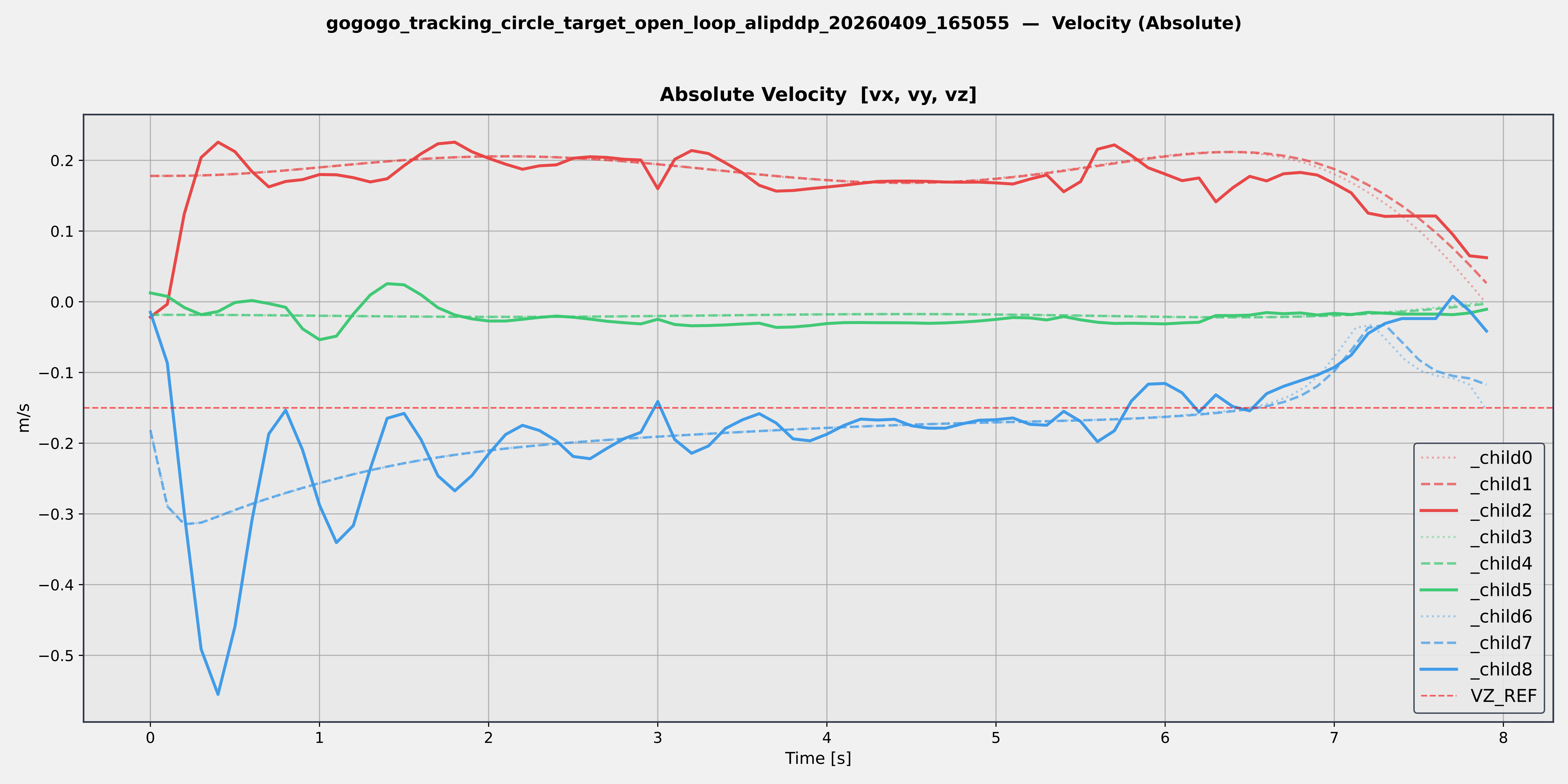
gogogo_tracking_circle_target_open_loop_alipddp_20260409_165055 — 3-D Trajectory (Absolute vs Relative)



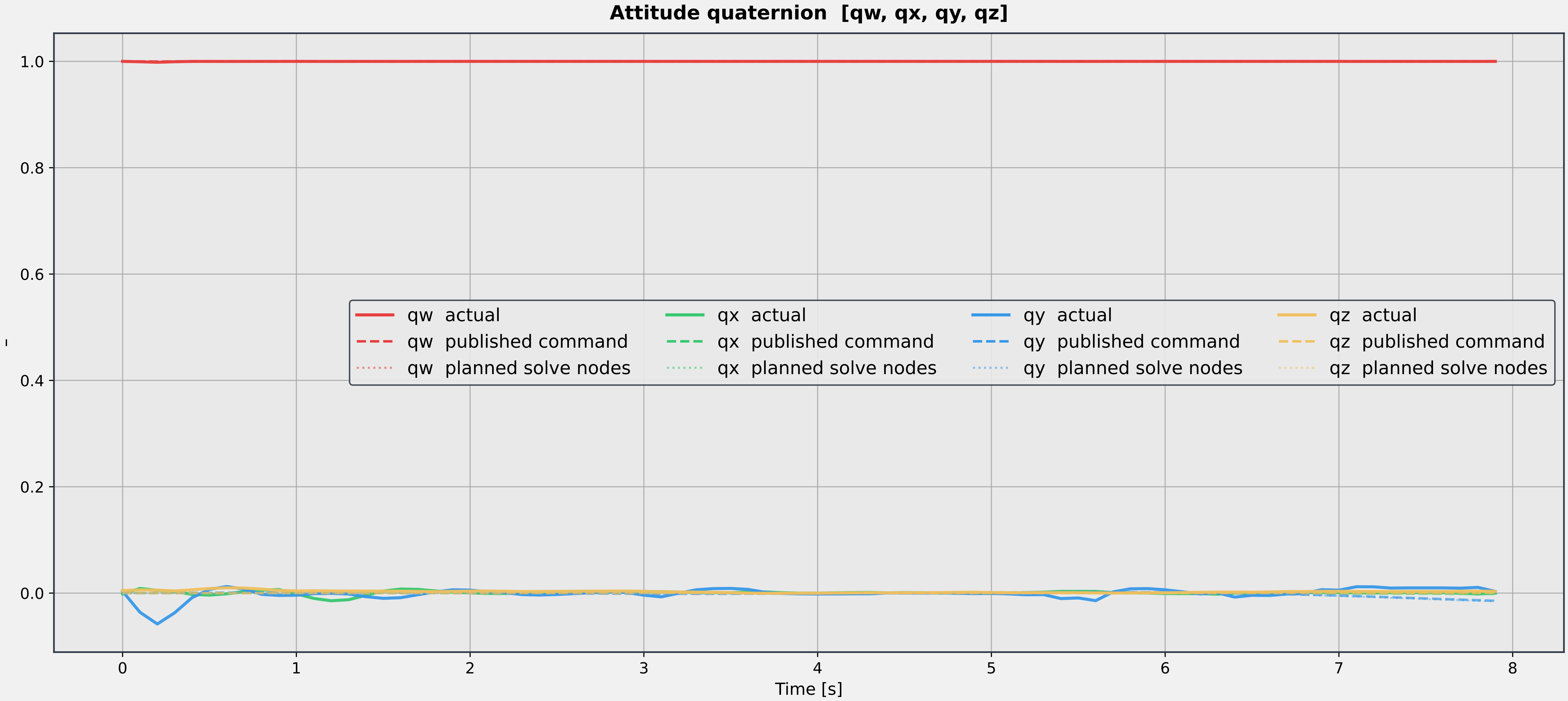
gogogo_tracking_circle_target_open_loop_alipddp_20260409_165055 — Position (Absolute)



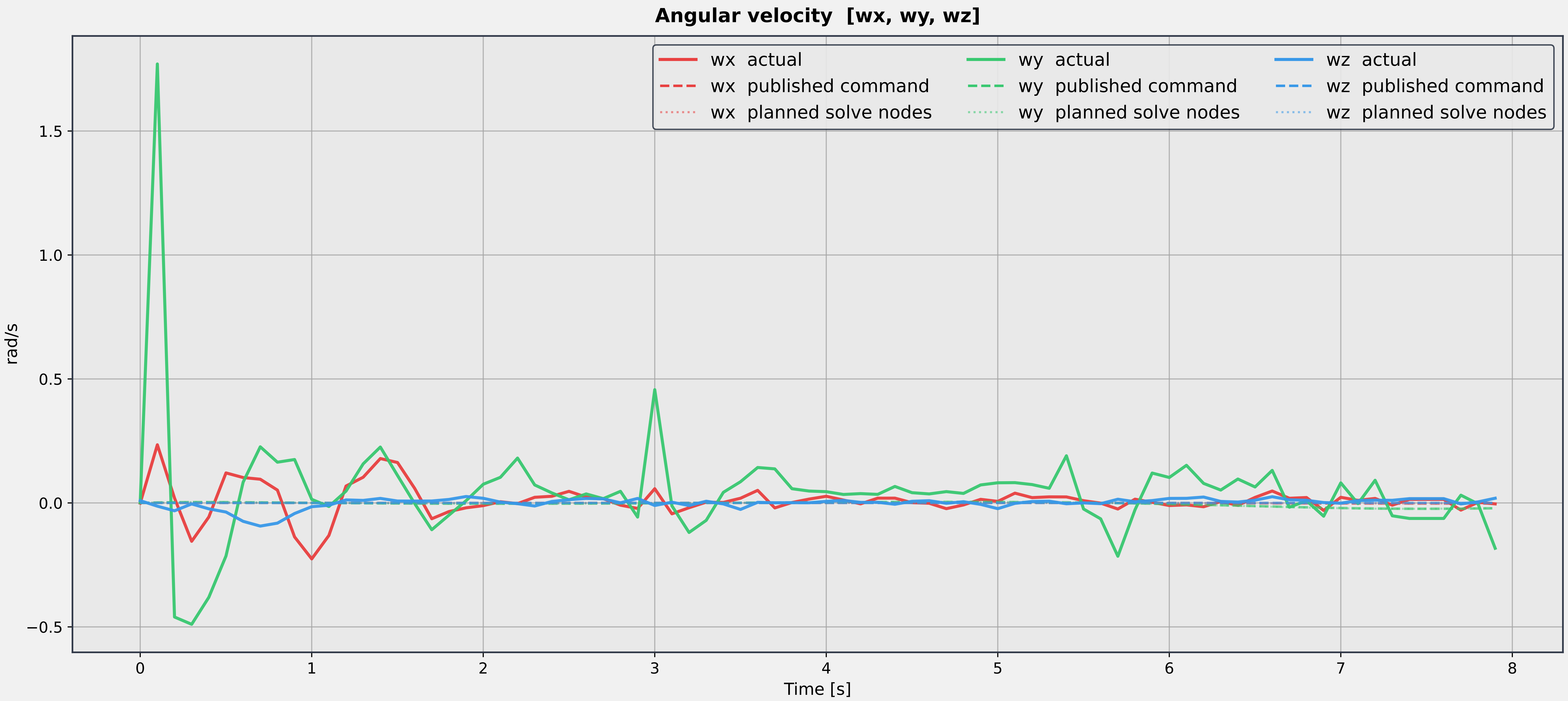


gogogo_tracking_circle_target_open_loop_alipddp_20260409_165055 — Velocity (Absolute)
Absolute Velocity [vx, vy, vz]
m/s
Time [s]
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
0
1
2
3
4
5
6
7
8
_child0
_child1
_child2
_child3
_child4
_child5
_child6
_child7
_child8
VZ_REF

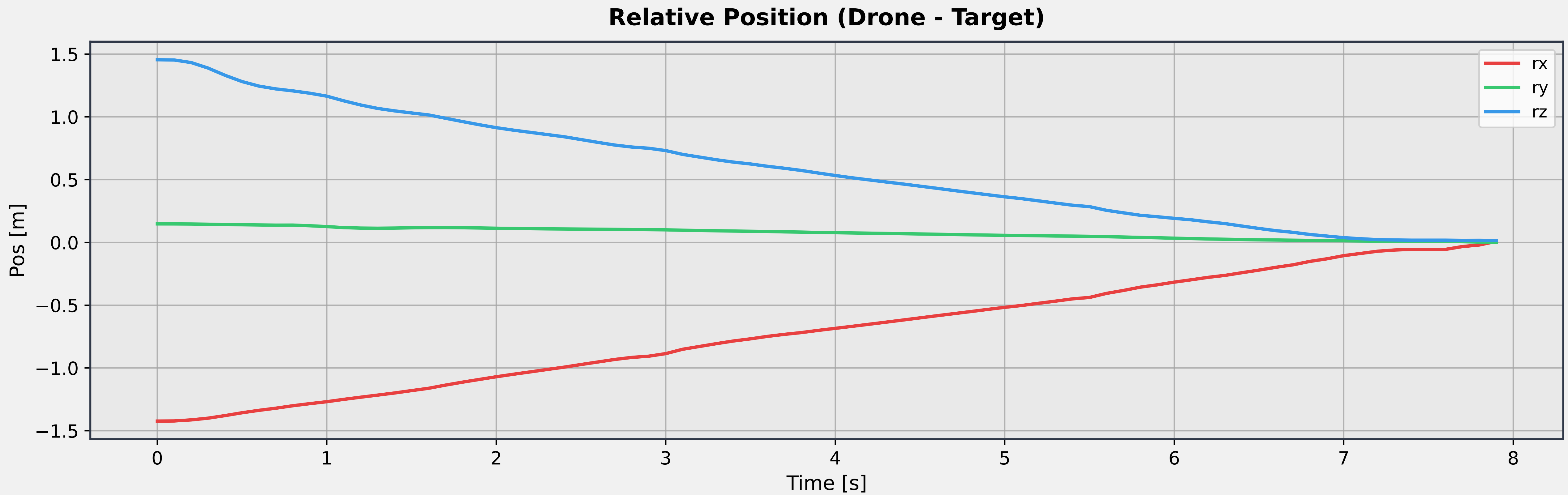
gogogo_tracking_circle_target_open_loop_alipddp_20260409_165055 — Quaternion

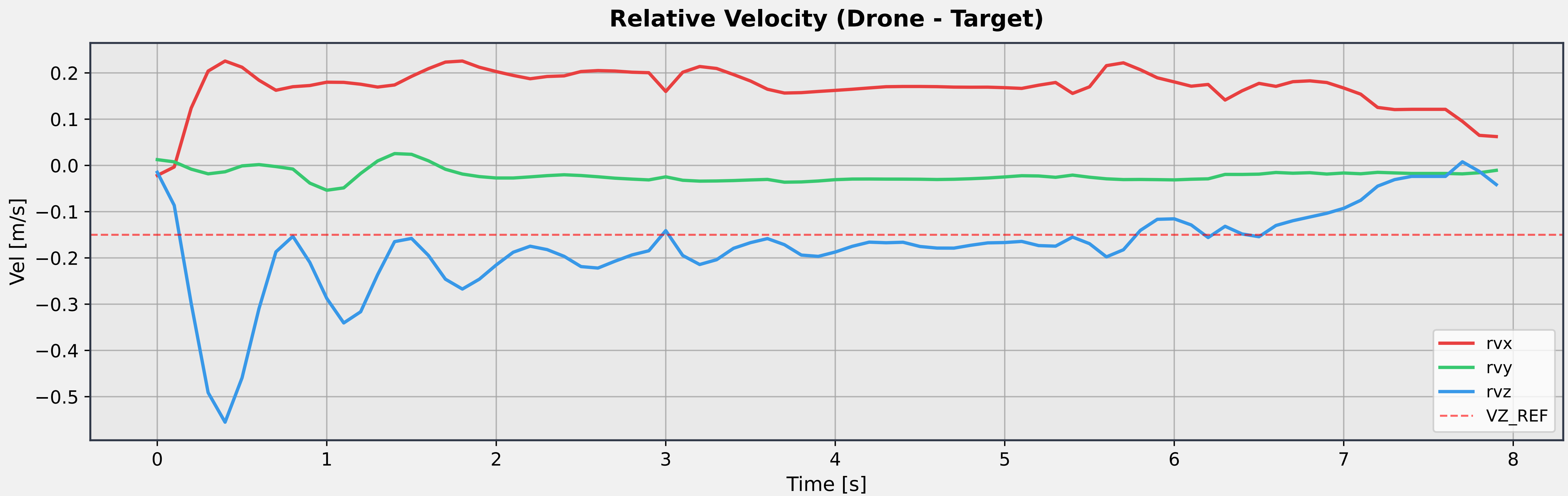


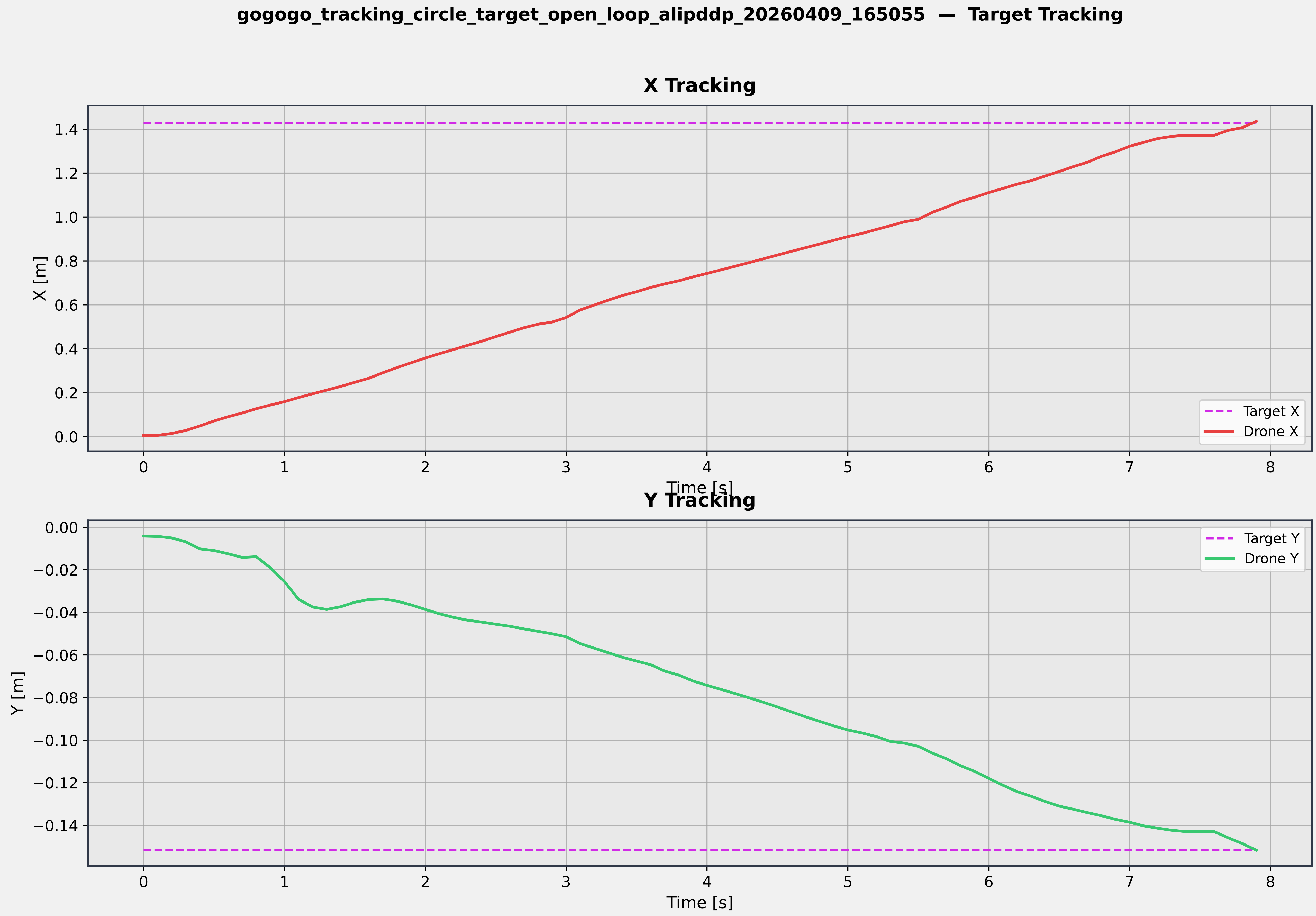
gogogo_tracking_circle_target_open_loop_alipddp_20260409_165055 — Angular Velocity

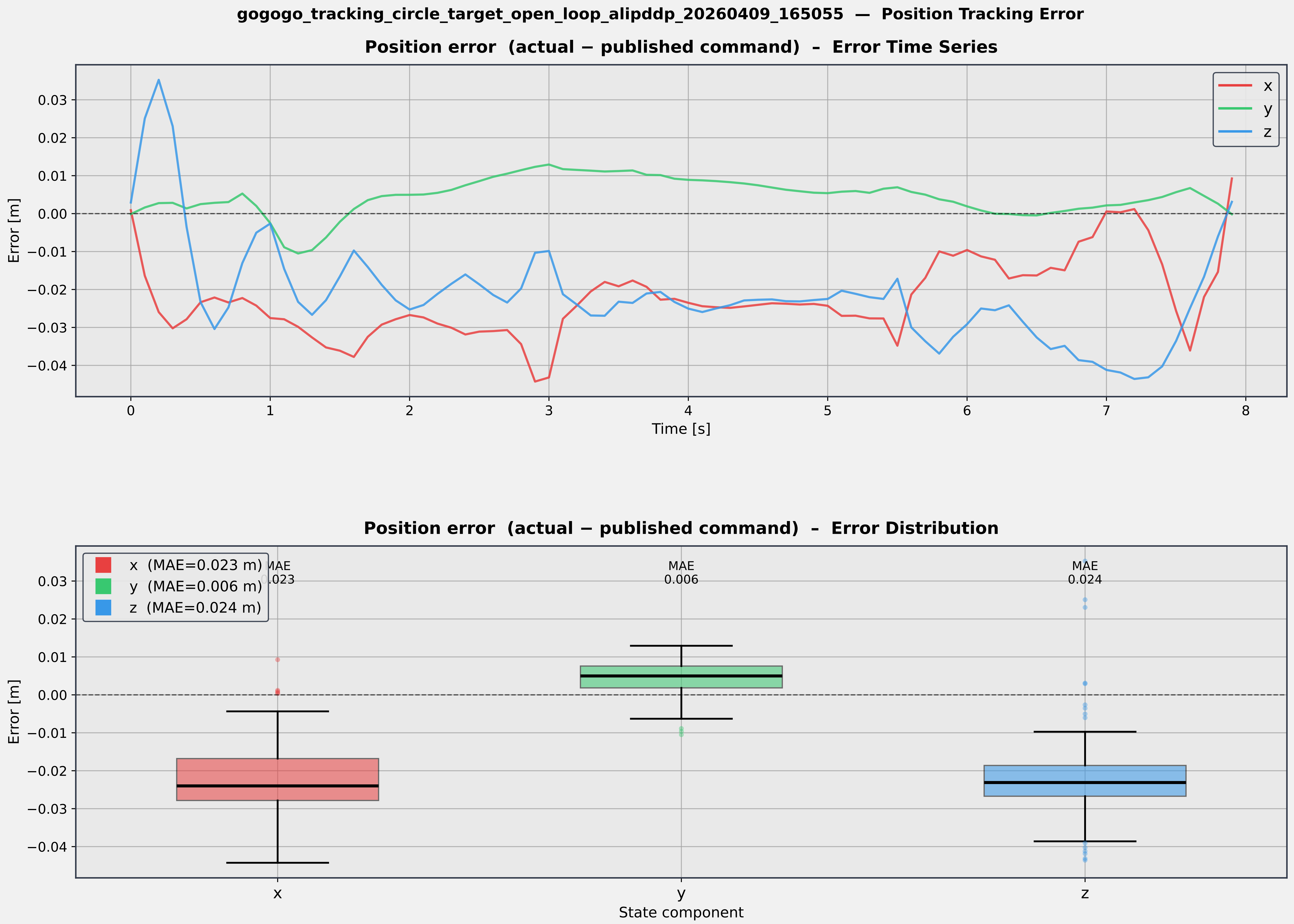


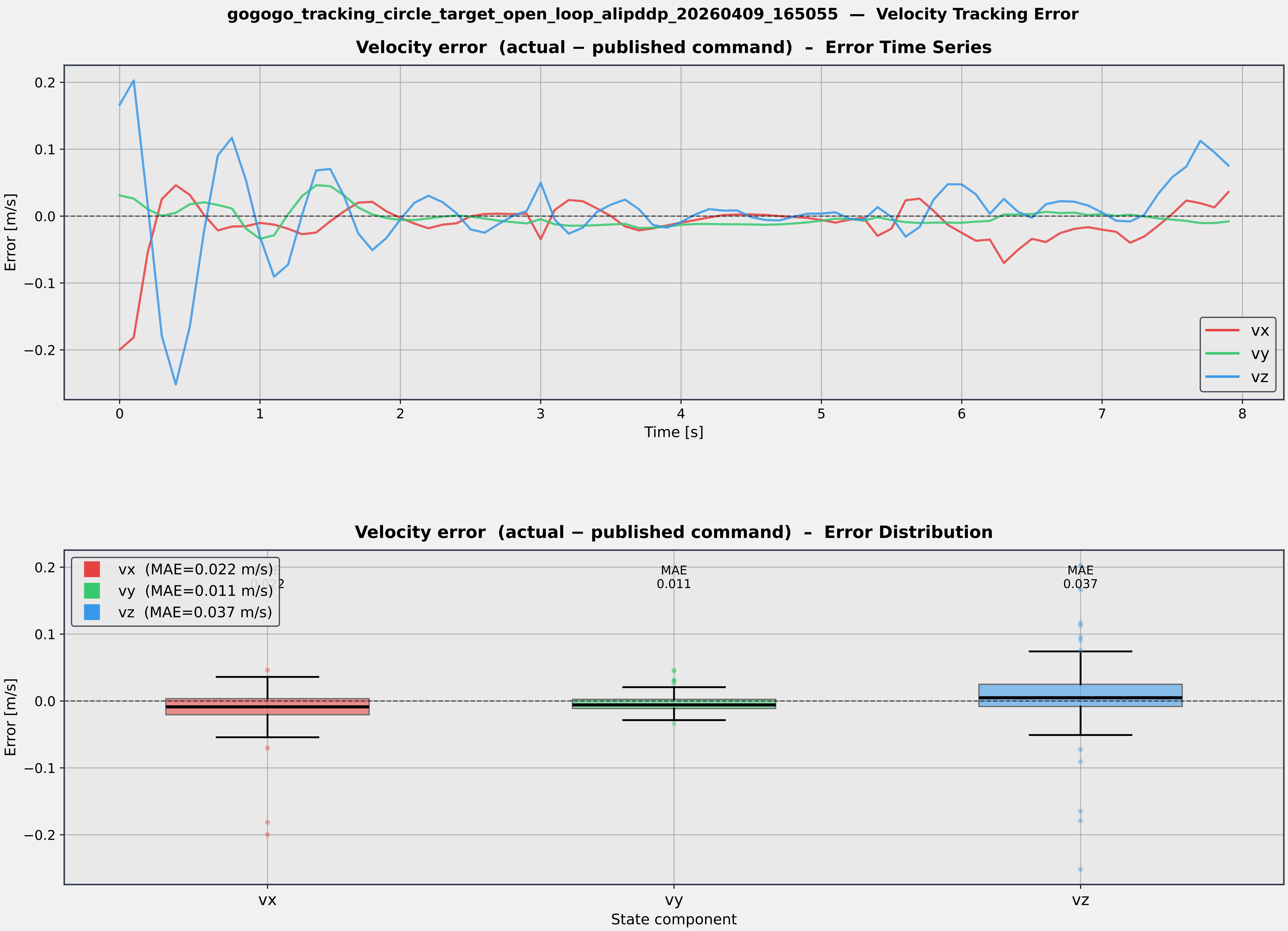
gogogo_tracking_circle_target_open_loop_alipddp_20260409_165055 — Relative States



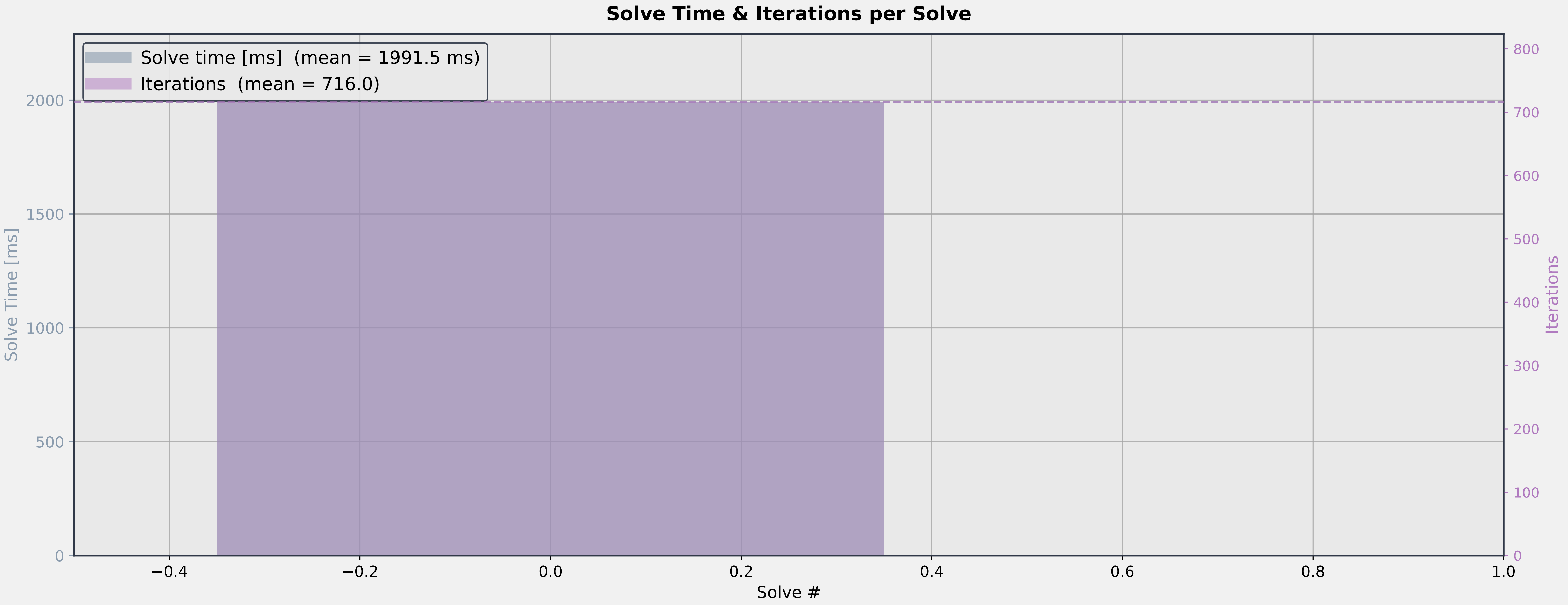


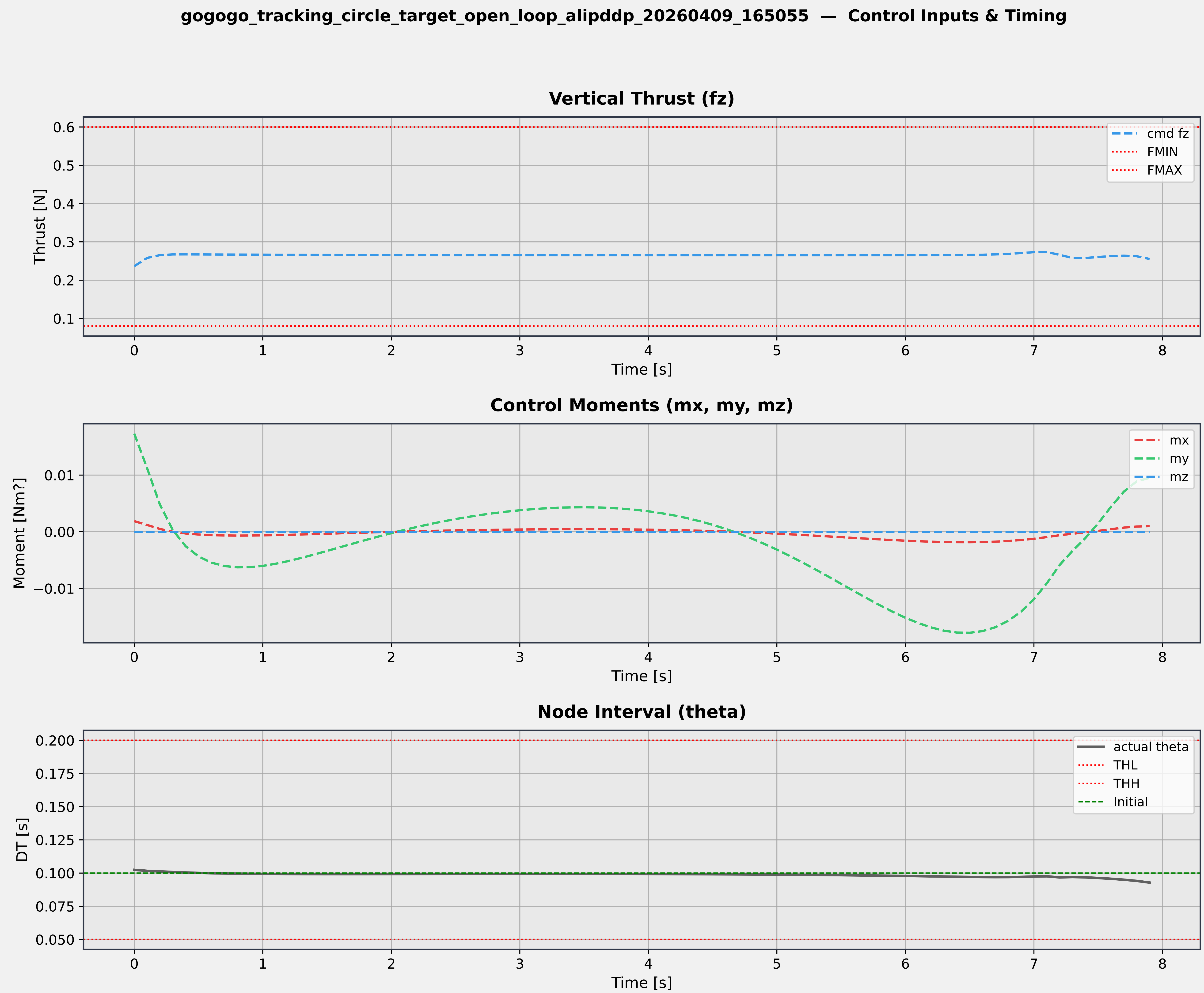

gogogo_tracking_circle_target_open_loop_alipddp_20260409_165055 — Target Tracking
X Tracking
X [m]
Time [s]
Target X
Drone X
Y Tracking
Y [m]
Time [s]
Target Y
Drone Y


gogogo_tracking_circle_target_open_loop_alipddp_20260409_165055 — Position Tracking Error
Position error (actual − published command) – Error Time Series
Error [m]
Time [s]
x
y
z
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
−0.04
0
1
2
3
4
5
6
7
8
Position error (actual − published command) – Error Distribution
x (MAE=0.023 m)
y (MAE=0.006 m)
z (MAE=0.024 m)
MAE
0.023
MAE
0.006
MAE
0.024
State component


gogogo_tracking_circle_target_open_loop_alipddp_20260409_165055 — Velocity Tracking Error
Velocity error (actual − published command) – Error Time Series
Error [m/s]
0.2
0.1
0.0
−0.1
−0.2
0
1
2
3
4
5
6
7
8
Time [s]
vx
vy
vz
Velocity error (actual − published command) – Error Distribution
vx (MAE=0.022 m/s)
vy (MAE=0.011 m/s)
vz (MAE=0.037 m/s)
MAE
0.011
MAE
0.037
Error [m/s]
0.2
0.1
0.0
−0.1
−0.2
vx
vy
vz
State component

gogogo_tracking_circle_target_open_loop_alipddp_20260409_165055 — Solver Performance




gogogo_tracking_circle_target_open_loop_alipddp_20260409_165055 — Control Inputs & Timing
Vertical Thrust (fz)
Thrust [N]
Time [s]
cmd fz
FMIN
FMAX
Control Moments (mx, my, mz)
Moment [Nm?]
Time [s]
mx
my
mz
Node Interval (theta)
DT [s]
Time [s]
actual theta
THL
THH
Initial